UNIVERSITY of BRADFORD

School of Computer Science, AI & Electronics

Portfolio:

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Supervisor: Professor Ciprian Daniel Neagu

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Chapter 1: Introduction

My portfolio serves as a comprehensive record of my professional development and project progress throughout the academic year. It provides an overview of the planning, execution, and reflection processes that have supported the successful completion of my final year project.

The portfolio is structured into several chapters. It begins with a detailed project plan, presented through Gantt charts, which outlines key tasks and timelines across both semesters, along with a discussion on any changes made during the project lifecycle. It then presents the chronological minutes of regular supervision meetings, highlighting achievements, ongoing challenges, and the objectives set during each session.

In addition, this document includes a Personal Development Plan that features an updated version of my CV and relevant career preparation materials, along with personal reflections on my career aspirations, such as planning to apply for postgraduate studies.

The final chapter showcases the poster prepared for the poster session, summarising the key elements of my project. This visual representation aims to communicate the project's purpose, development process, and outcomes effectively.

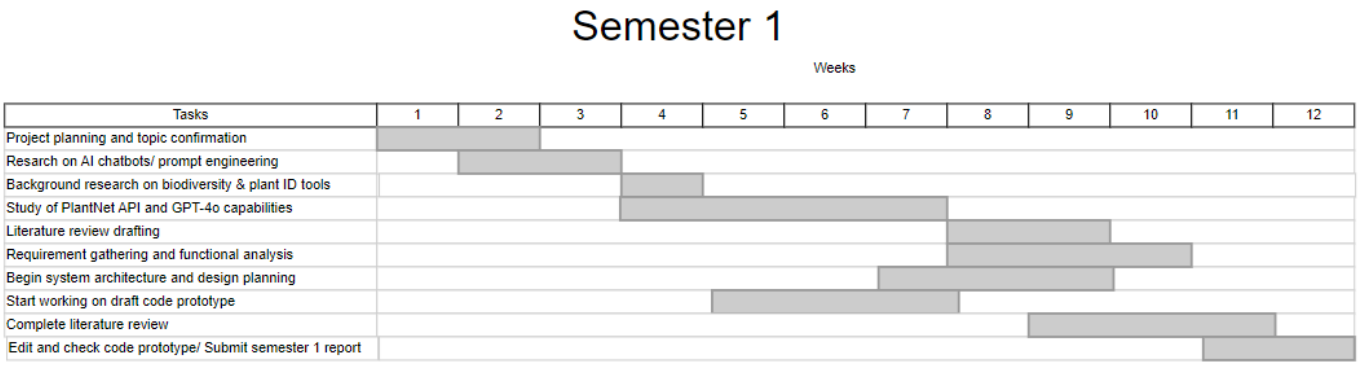
Together, these components provide evidence of my continuous development, engagement with the project, and preparedness for professional practice.

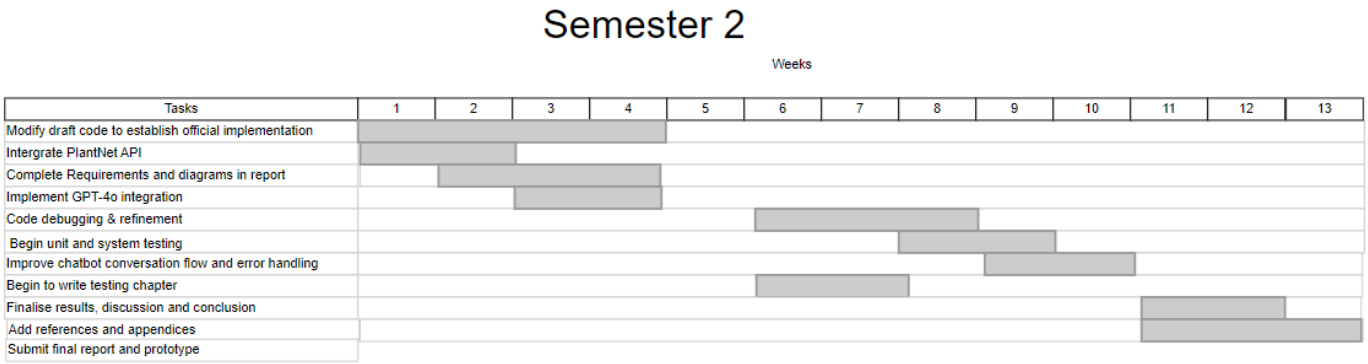
Chapter 2: Plan

The planning and organisation of the project were essential to ensure the successful completion of each development phase. Two detailed Gantt charts was created to outline all key tasks, starting from the initial stages of research and requirement gathering to the final phases of implementation, testing, and documentation. The project was divided across two academic semesters, with each week assigned a specific focus. This structured timeline helped monitor progress and allowed flexibility for adjustments where necessary.

During Semester 1, the emphasis was placed on foundational tasks, including exploring prompt engineering, studying image-processing tools, understanding GPT-4o capabilities, and reviewing related biodiversity technologies. The planning phase also included formulating clear project objectives and analysing user and system requirements. This early preparation ensured that a strong conceptual framework was in place for development in the following term.

Semester 2 focused primarily on the implementation and testing stages. Which involved refining prompts, debugging issues, conducting unit and system testing, and preparing the final report. The Gantt chart not only provided a visual representation of project progression but also enabled the tracking of dependencies between tasks, helping maintain momentum throughout the project lifecycle.





Throughout the course of the project, several adjustments were made to the original Gantt chart and project plan to accommodate unforeseen challenges and evolving project requirements. In the early stages, tasks such as initial research, prompt engineering exploration, and requirements analysis were completed on schedule. However, as the development progressed into Semester 2, certain areas, particularly the integration of GPT-4o, required more time than originally planned. Understanding the GPT-4o model's behaviour, especially for generating accurate and relevant ecological insights, proved more complex than anticipated. As a result, the time allocated for prompt engineering and API integration was extended by two weeks, delaying the start of some testing phases.

Despite these delays, other tasks were adjusted accordingly to ensure the project remained on track. For example, documentation and report writing were carried out in parallel with testing and debugging, which helped mitigate the impact of earlier setbacks. Additionally, the initial plan did not account for the full scope of system-level testing, which was later added after receiving supervisor feedback. These changes, while shifting the overall timeline, allowed for a more refined and user-focused system to emerge. This flexible approach to planning and iterative development ensured the final product met its objectives while also improving the robustness and usability of the chatbot.

Chapter 3: Supervision meeting minutes

Regular supervision meetings were held throughout the project to ensure steady progress, receive feedback, and align the work with the project objectives. These meetings typically took place once a week, depending on the stage of the project. Below is a chronological summary of the key meetings, including progress updates, upcoming goals, and any challenges encountered during development.

The formal start of my final year project began on the 25th of September 2024, during Week 1 of the academic term. At this stage, I opted for the NCD2 project stream titled *"Responsible Generative AI Chatbots for Logical Problem Solving and Blended Academic Research Delivery"*. This topic strongly resonated with both my interests in artificial intelligence and my technical strengths in software engineering. I informed my assigned supervisor, Professor Daniel, of my decision to pursue this stream. Once the choice was officially confirmed and recorded, I was encouraged to start my preliminary research. As a foundational step, I was given access to a relevant O'Reilly publication which included a dedicated section on prompt engineering. This served as an excellent starting point to build a comprehensive understanding of how generative AI systems, can be shaped and optimised through carefully crafted prompts. The book also provided insights into responsible use, bias mitigation, and methods for enhancing the relevance and accuracy of chatbot outputs, elements that would later form the core principles of my own project.

In parallel with the early research phase, I took part in a number of collaborative academic sessions that helped shape the direction of my project. Throughout October and into November 2024, I participated in a dedicated series of BNG (Biodiversity Net Gain) group article/processing meetings. These sessions were held weekly, specifically on the 14th (34 minutes), 21st (30 minutes), and 28th (55 minutes) of October, followed by the 4th (60 minutes), 11th (59 minutes), and 18th (57 minutes) of November. These group meetings brought together fellow FYP tutees and an MSc tutee, all of whom were exploring related themes within the BNG stream. During these collaborative gatherings, we explored the core academic and policy frameworks surrounding biodiversity net gain in the UK, critically evaluated articles and current debates, and discussed how ideas could align with broader environmental and conservation goals. The group setting offered not only a support network but also a platform for cross-pollination of ideas. From defining individual project scopes to locating datasets and useful resources, these sessions significantly influenced the formulation of my objectives.

Concurrently, I attended weekly FYP tutorial meetings that spanned the months of October through December 2024. These meetings occurred on the 8th (59 minutes), 15th (59 minutes), 22nd (35 minutes), and 30th (48 minutes) of October; the 5th (60 minutes), 12th (31 minutes), 19th (41 minutes), and 26th (46 minutes) of November; and the 3rd (39 minutes) of December. These tutorials were essential to the development of the technical backbone of the project. During the earlier sessions, I confirmed my specific focus within the NCD2 stream—namely, the combination of prompt engineering and image processing for BNG. As the meetings progressed, we explored and discussed relevant technologies such as Claude, GPT-4o, and various image processing techniques that I could use in my project. These sessions not only helped refine the scope of my technical implementation but also provided an opportunity to clarify submission timelines and expectations for

various project milestones, particularly Submission 1. Weekly progress updates were also a key part of these tutorials. I regularly shared insights from my literature review, outlined the draft requirements specification, and presented early-stage prototypes of the codebase. These meetings ensured that development and documentation moved in tandem, fostering a holistic and well-paced workflow throughout the term.

As the academic year progressed into Semester 2, my participation in structured meetings continued. On the 4th of February 2025, I attended a regular FYP drop-in session (26 minutes), which served as a touchpoint for feedback. These informal check-ins allowed me to raise specific concerns and get guidance on immediate challenges in both development and documentation. Finally, during the crucial final phase of the project in early April 2025. I sent a full draft of my dissertation and working codebase to Professor Daniel via email. He responded with detailed and constructive feedback which enabled me to polish the final report and refine explanations to ensure clarity and completeness. This last round of communication was instrumental in finalising a well-rounded, academically thorough project that reflected both the initial aims and the iterative progress made throughout the year. The structured supervision, consistent feedback, and collaborative academic setting provided the ideal environment to realise the goals of my final year project in a focused and professional manner.

3.1 Supervision meeting log

25 September 2024 – Week 1 (Initial Project Setup)
Duration: N/A
Discussion: I informed my supervisor, Professor Daniel, of my decision to select the NCD2 project stream titled *"Responsible Generative AI Chatbots for Logical Problem Solving and Blended Academic Research Delivery."* Once the selection was confirmed and recorded, I was recommended a relevant O'Reilly book with sections on prompt engineering to begin early research.
Outcome: Project choice confirmed, and initial reading assigned.
Next Steps: Begin report introduction focusing on prompt engineering and generative AI foundations.

14 October 2024 – Week 4 (BNG Group Meeting)
Duration: 34 minutes
Discussion: Participated in group meeting with fellow FYP and MSc students to explore the BNG stream. Discussion focused on current policy background, research direction, and defining individual project scopes.
Outcome: Better understanding of BNG context; began aligning AI project with biodiversity conservation objectives.
Next Steps: Identify relevant tools and APIs for flora detection.

21 October 2024 – Week 5 (BNG Group Meeting)
Duration: 30 minutes
Discussion: Continued discussion on BNG academic resources and scope alignment.

Outcome: Clarified relevance of image processing for biodiversity; established clearer direction for system functionality.
Next Steps: Explore PlantNet API and confirm feasibility.

28 October 2024 – Week 6 (BNG Group Meeting)
Duration: 55 minutes
Discussion: Reviewed biodiversity data sources and began discussing individual project implementation plans.
Outcome: Decided to use PlantNet API and integrate GPT-4o for chatbot support.
Next Steps: Begin early system design.

4 November 2024 – Week 7 (BNG Group Meeting)
Duration: 60 minutes
Discussion: Developed initial project ideas reviewed feasibility of data pipeline using images.
Outcome: Finalised concept of AI chatbot using plant images and prompt engineering.
Next Steps: Design diagrams and data flow models.

11 November 2024 – Week 8 (BNG Group Meeting)
Duration: 59 minutes
Discussion: Identified image data availability.
Outcome: Defined output formats and refined system requirements.
Next Steps: Start implementation of base Python script.

18 November 2024 – Week 9 (BNG Group Meeting)
Duration: 57 minutes
Discussion: General BNG discussions.
Outcome: Project scope finalised, initial diagrams created.
Next Steps: Continue development, begin testing prototype.

8 October – 3 December 2024 (Weekly FYP Tutorials)
Dates & Durations:

- 8 Oct (59m), 15 Oct (59m), 22 Oct (35m), 30 Oct (48m)
- 5 Nov (60m), 12 Nov (31m), 19 Nov (41m), 26 Nov (46m)
- 3 Dec (39m)

Discussion: Weekly progress updates and technical discussions on generative AI, GPT-4o, Claude, and image processing. Confirmed decision to work on prompt engineering for flora recognition. Discussed requirements and started initial code prototyping. Submission 1 documentation milestones tracked.
Outcome: Clear direction and iterative progress on literature review, requirements analysis, and code development.
Next Steps: Complete prototype and test, finalise requirement specs and initial draft of report.

4 February 2025 – Semester 2 Begins (FYP Drop-in)
Duration: 26 minutes
Discussion: Attended regular drop-in to address development progress and get feedback on system design and testing phase.
Outcome: Validated code direction; planned documentation.
Next Steps: Write testing report and refine chatbot logic.

Early April 2025 – Final Submission Review
Discussion: Emailed final dissertation and full working code to Professor Daniel for feedback.
Outcome: Supervisor provided constructive final feedback for improvements in documentation.
Next Steps: Incorporated feedback, performed final edits, and submitted the final report, portfolio and demonstration recording.

The supervision meetings were essential in guiding the project from its initial idea to final completion. Regular sessions helped refine the research focus, resolve technical challenges, and ensure steady progress. They also provided useful feedback on documentation and implementation, improving the final output. These meetings supported both academic development and practical skills like problem-solving and time management, contributing significantly to the success of the project

Chapter 4: Personal Development Plan

As I approach the end of my undergraduate studies, I have taken time to reflect on my personal and professional development over the past few years. This chapter outlines the steps I have taken to prepare for life after graduation, including the development of my CV and the exploration of both career and academic opportunities. While I have actively prepared for potential employment, I am also planning to apply for postgraduate studies in Artificial Intelligence. My interest in AI has grown steadily throughout my degree, particularly through coursework and projects involving machine learning and intelligent systems. Pursuing postgraduate education will allow me to deepen my understanding of AI technologies, engage in research, and better position myself for a career at the forefront of innovation. The following sections include the documents I have prepared and reflections on my career planning journey.

Below is my most up-to-date CV, which highlights my academic background, technical skills, and relevant experiences:

Appendix 1:

Mishel Rehan

Currently Bradford-based but flexible on location.
M: 7305 327251 E: mishelr999@outlook.com

EDUCATION AND QUALIFICATIONS

University of Bradford **2022 – 2025 BEng (Hons) Software Engineering**
Predicted: 1st Class.

- ***University Projects*** – Final Year Project: 'Generative Artificial Intelligence Chatbot Using Prompt Engineering for Biodiversity Net Gain'. Focused on building an intelligent plant identification chatbot leveraging GPT-4o, combining software engineering principles, AI integration, and ecological awareness.

- ***Key Modules (Year 3)*** – *Final Year Project*, Software Systems Design and Testing, Machine Learning, Large Scale Data Driven Applications.

- ***Key Modules (Year 1 to Year 2)*** – Yr2: Artificial Intelligence, Yr2: Data Structures and Algorithms, Yr2: Database Systems, Yr2: System Security Management, Yr1: Fundamentals of Programming, Yr1: Mathematics for Computing, Yr1: Technical and Professional Skills.

St Benedict's Sixth Form, Bradford **2020-2022**
GCE A Levels: IT (M), Criminology (C), Business Studies (M)

St Bede's & St Joseph's Catholic College, Braford **2018-2020**
GCSEs: 9 subjects (2As, 3Bs and Cs) including Mathematics, English, and French.

TECHNICAL / SOFTWARE AND LABORATORY SKILLS

- Possess extensive knowledge of the key fundamentals in Java, SQL, PHP, Python, CSS, HTML, JavaScript which are frequently being expanded on.
- Experience in MySQL, PhpMyAdmin and MongoDB.
- Proficient in using all Microsoft Word Packages.

WORK EXPERIENCE

Sales Assistant June 2021 – Present Day
BSX, Bradford, West Yorkshire

- Responsible for sorting and pricing stock to ensure efficient product turnover.
- Delivered excellent customer service through clear and friendly communication.

- Handled transactions accurately and confidently, including cash and card payments.
- Managed and updated the company website, including uploading new stock and maintaining product information.

Self-Employed E-commerce Seller September 2022 - Present Day

eBay & Vinted

- Operating a successful resale business across eBay and Vinted, managing end-to-end operations
- Handled inventory management, product sourcing, pricing, and listing creation
- Provided excellent customer service, resolving issues promptly to maintain high satisfaction and positive feedback
- Continuously managing order fulfilment, including packaging and shipping logistics, and maintained supply levels for smooth operations

WORK EXPERIENCE SKILLS

Customer Service & Communication

- Gained strong interpersonal skills by assisting a wide range of customers in-store and online.
- Resolved issues effectively and professionally, maintaining high customer satisfaction and loyalty.
- Adapted communication style to suit different customer needs and platforms.

Sales & Product Knowledge

- Developed keen attention to detail in pricing, product presentation, and stock rotation
- Identified trends and popular items to optimize listings and boost sales

E-commerce Management

- Ran all aspects of a resale business on eBay and Vinted, including sourcing, listing, marketing, and fulfilment
- Managed and updated product inventory and ensured timely shipping and accurate order processing

Website & Digital Tools

- Gained hands-on experience updating business websites and product pages
- Confident using digital platforms, including eBay, Vinted, and basic content management tools

Money Handling & Financial Awareness

- Accurately handled cash and card payments in a retail setting
- Managed budgeting, expenses, and profits for an independent business

Organisational & Time Management

- Balanced multiple tasks and priorities between store responsibilities, business operations and studies
- Maintained consistent workflow and order tracking to meet customer expectations and deadlines

LEISURE INTERESTS

- Love collecting retro and rare toys, games, and old-school consoles – always on the hunt for hidden gems
- Keen horse rider – enjoy being outdoors and spending time with animals
- Retired Jujitsu purple belt – trained for years and still carry that discipline and focus into everyday life

Looking ahead, I plan to apply for postgraduate study in Artificial Intelligence and Machine Learning at the University of Bradford, which aligns closely with my academic interests and long-term career goals, offering an in-depth exploration of cutting-edge AI technologies, including neural networks, deep learning, and data-driven decision-making. The university's strong emphasis on practical, hands-on learning and its reputation for research in intelligent systems make it an ideal environment for me to further develop both my theoretical knowledge and technical skills. I believe that completing this course will provide me with the advanced expertise needed to contribute meaningfully to the AI field and pursue a career in innovative, high-impact technology roles.

Appendix B

Artificial Intelligence and Machine Learning MSc

Apply for Artificial Intelligence and Machine Learning MSc at the University of Bradford.

Suitable for: UK and International applicants

Attendance: Full-time, Part-time

Technology and Artificial Intelligence Law LLM

Apply for Technology and Artificial Intelligence Law LLM at the University of Bradford.

Suitable for: UK and International applicants

Attendance: Full-time, Part-time

Applied Artificial Intelligence and Data Analytics MSc

Apply for Applied Artificial Intelligence and Data Analytics MSc at the University of Bradford.

Suitable for: UK and International applicants

Attendance: Full-time, Part-time

Artificial Intelligence and Space Technology MSc

Apply for Artificial Intelligence and Space Technology MSc at the University of Bradford.

Suitable for: UK and International applicants

Attendance: Full-time, Part-time

4.1 Reflections on career prospects

Reflecting on my career prospects, I feel both optimistic and motivated by the rapid growth of the Artificial Intelligence industry and the increasing demand for professionals skilled in machine learning, data analysis, and intelligent system design. My academic background, combined with my growing interest in AI, positions me well to enter a field that is not only technically challenging but also socially impactful. Whether I pursue a career in AI research, software development, or applied machine learning, I am confident that the skills and knowledge I have developed so far, along with further specialisation through postgraduate study, will open diverse and exciting opportunities. I also recognise the importance of ongoing learning and adaptability, particularly in a field as fast-evolving as AI.

Chapter 5: The Showcase Poster

UNIVERSITY of BRADFORD
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AI-driven Flora ID Chatbot

Identify plant species using AI to support biodiversity conservation.

- Integrated PlantNet API for image-based plant recognition.
- Used GPT-4o for generating ecological insights
- .Designed a chatbot interface for user interaction.

```
🌿 BNGbot - Biodiversity Assistant 🌿
Enter the file path for your plant image: Images\plant1.jpeg

✅ Identified species: Anthoxanthum odoratum

🧠 Generating ecological insights...

📋 Ecological Summary:
 ### Ecological Summary of *Anthoxanthum odoratum* (Sweet Vernal Grass)

#### Habitat:
*Anthoxanthum odoratum* is a perennial grass species native to Europe and parts of Asia but has been widely intro
duced to North America, New Zealand, and Australia. It is commonly found in meadows, pastures, grasslands, woodla
nd edges, and roadsides. Sweet vernal grass thrives in a variety of soil types, including acidic, neutral, and ca
lcareous soils, and prefers well-drained locations with moderate moisture. It is tolerant of low nutrient conditi
ons and often establishes itself in disturbed areas.
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- AI improves accessibility to plant identification.
- Supports BNG (Biodiversity Net Gain) initiatives.
- Effective use of prompt engineering and APIs.
- Future scope includes adding more species and mobile deployment.

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Software Engineering
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